

Appendix 1

Client-Efficient Accuracy and Minority Robustness

Table 1 provides a detailed comparison of MiB-SCFL against representative FL baselines under non-IID settings across vision, sensor, and language datasets. In addition to total accuracy/loss, minority accuracy/loss, and F1-score, the table reports model parameters, FLOPs per sample, server-side parameters, client-side cost per round, and client-side trainable parameters per round. This extended view is important because, in practical federated learning, the main resource bottleneck is usually on the client side rather than the server. While the server can often store and coordinate multiple models, clients are typically constrained by computation, memory, and energy. Therefore, we emphasize client cost per round and client trainable parameters per round as the main efficiency criteria.

The results show that MiB-SCFL achieves a favorable balance between predictive performance and client-side efficiency. Unlike multi-model or clustering-based baselines such as B-CFL, FedRFC, FedSoft, MCFL, and SC-FL, which may require clients to train or process multiple full models per round, MiB-SCFL keeps the client-side footprint close to a single model. For example, on MNIST, HAR, and PAMAP2, MiB-SCFL uses approximately $0.99\times$ full client cost, whereas FedRFC and SC-FL require up to $3\times$ full cost. On Shakespeare, where the sequence-modeling task is more challenging, MiB-SCFL increases the client cost moderately to about $1.15\times$ full, still remaining substantially below the $3\times$ cost required by stronger multi-model baselines such as FedSoft and FedRFC.

Despite this lightweight client-side design, MiB-SCFL consistently improves minority robustness. On MNIST, MiB-SCFL achieves the best minority accuracy of 97.49%, outperforming FedAvg (95.76%), FedProx (95.71%), FedSoft (95.86%), and FedRFC (96.96%). On CIFAR-10, MiB-SCFL also obtains the best minority accuracy of 54.84%, improving over FedAvg (49.54%), FedProx (48.55%), pFedCAM (50.24%), and FedRFC (44.15%). On HAR, MiB-SCFL reaches 89.09% minority accuracy, nearly matching FedRFC (89.11%), while using about one-third of its client-side cost. On PAMAP2, MiB-SCFL remains competitive with 93.12% minority accuracy and achieves the best F1-score of 94.83%. These results indicate that MiB-SCFL improves the performance of under-represented groups without relying on a larger client-side model footprint.

MiB-SCFL also provides strong overall accuracy. It achieves the best total accuracy on CIFAR-10 (68.13%), clearly outperforming FedAvg (62.69%), FedProx (62.74%), pFedCAM (62.86%), FedSoft (59.85%), and FedRFC (59.13%). On PAMAP2, MiB-SCFL obtains the highest total accuracy of 96.77% and the best F1-score of 94.83%. On HAR, it reaches 92.51% total accuracy, remaining close to FedRFC (93.79%), but with a much smaller client-side burden. These results show that the gains of MiB-SCFL are not limited to minority groups; the method also preserves, and often improves, general predictive performance.

The comparison further highlights an important limitation of several strong baselines. Methods such as FedRFC and FedSoft can perform well in some cases, but their improvements often come with substantially higher client-side cost. For instance, FedRFC uses $3\times$ full client cost on MNIST, CIFAR-10, HAR, PAMAP2, and Shakespeare, while MiB-SCFL remains close to $1\times$ on the first four datasets and only $1.15\times$ on Shakespeare. Moreover, FedRFC degrades sharply on Shakespeare, reaching only 29.23% total accuracy and 14.29% minority accuracy, whereas MiB-SCFL maintains 50.13% total accuracy and 22.85% minority accuracy. FedSoft achieves the strongest Shakespeare accuracy, but it requires $3\times$ full client cost, compared with only $1.15\times$ for MiB-SCFL.

Overall, the appendix results strengthen the main conclusion: MiB-SCFL does not achieve its gains by simply increasing the burden on clients. Instead, it uses compact cluster-specific subnetworks to obtain the benefits of specialization while keeping client-side training close to the cost of a single global model. This leads to a robust and practical accuracy–efficiency trade-off: MiB-SCFL improves minority performance, remains competitive in total accuracy, and substantially reduces client-side cost compared with heavier multi-model baselines. These properties make MiB-SCFL particularly suitable for realistic non-IID federated learning scenarios where client devices are resource-constrained but minority robustness remains essential.

Table 1: Performance and client-side efficiency under non-IID settings across vision, sensor, and language datasets. We report total accuracy/loss, minority accuracy/loss, F1-score (%), model parameters, FLOPs/sample, server parameters, and client-side cost/parameters per round. Client-side cost is emphasized because clients are the primary resource-constrained devices in FL.

Set.	Method	Total Acc./Loss	Minority Acc./Loss	F1 (%)	Param./Model	FLOPs/Sample	Server Param.	Client Cost/Round	Client Param./Round
MNIST & $\epsilon = 0.05$	FedAvg	96.99±0.493/0.0933±0.173	95.76±0.576/0.1287±0.379	96.96±13.10%	421,738	4,241,152	421,738	1× full	421,738
	FedProx	96.94±0.483/0.0941±0.121	95.71±0.586/0.1272±0.359	96.90±16.20%	421,738	4,241,152	421,738	1× full	421,738
	CFL	68.30±0.679/1.1748±0.421	92.24±0.669/0.7970±0.524	91.92±29.70%	421,738	4,241,152	1,265,214	1× full	421,738
	B-CFL	87.85±0.566/0.3529±0.317	71.83±0.597/0.5591±0.348	87.19±20.40%	421,738	4,241,152	1,265,214	3× full	1,265,214
	FedMTL	40.27±0.648/1.8675±0.431	31.35±0.690/1.9744±0.452	28.39±28.60%	421,738	4,241,152	6,326,070	1× full	421,738
	FedRFC	97.23±0.442/0.1229±0.183	96.96±0.617/0.1241±0.400	97.19±10.00%	421,738	4,241,152	1,265,214	3× full	1,265,214
	FedSoft	97.10±0.462/0.0899±0.111	95.86±0.628/0.1255±0.390	97.07±14.20%	421,738	4,241,152	1,265,214	3× full	1,265,214
	FedSPD	84.41±0.555/0.4687±0.214	84.61±0.700/0.4074±0.411	81.79±22.40%	421,738	4,241,152	1,265,214	1× full	421,738
	MCFL	88.47±0.535/0.4133±0.307	73.66±0.638/0.6713±0.369	87.94±24.50%	421,738	4,241,152	1,265,214	2× full	843,476
	SC-FL	89.45±0.514/0.4104±0.255	75.97±0.607/0.5712±0.338	89.16±23.50%	421,738	4,241,152	1,265,214	3× full	1,265,214
	pFedCAM	88.02±0.545/0.3884±0.276	79.73±0.659/0.5631±0.328	87.83±19.30%	421,738	4,241,152	1,265,214	1× full	421,738
	MiB-SCFL	97.13±0.473/0.1086±0.152	97.49±0.504/0.1137±0.266	97.06±9.00%	421,738	4,241,152	1,265,214	$\approx 0.99\times$ full	$\approx 421,728$
CIFAR-10 & $\epsilon = 0.5$	FedAvg	62.69±1.450/1.1272±0.839	49.54±1.664/1.4857±1.114	56.18±62.50%	716,778	43,485,696	716,778	1× full	716,778
	FedProx	62.74±1.542/1.1320±0.778	48.55±1.633/1.5249±1.083	60.51±59.40%	716,778	43,485,696	716,778	1× full	716,778
	CFL	41.19±1.939/1.9395±1.267	10.64±1.908/3.1257±1.358	44.92±68.60%	716,778	43,485,696	2,150,334	1× full	716,778
	B-CFL	58.56±1.481/1.2355±0.900	45.48±1.878/1.4121±1.236	58.33±50.30%	716,778	43,485,696	2,150,334	3× full	2,150,334
	FedMTL	17.63±1.847/2.2182±1.144	52.10±1.725/2.3919±1.206	12.09±56.40%	716,778	43,485,696	10,751,670	1× full	716,778
	FedRFC	59.13±1.419/1.2034±0.717	44.15±1.969/1.4255±1.297	58.39±41.10%	716,778	43,485,696	2,150,334	3× full	2,150,334
	FedSoft	59.85±1.572/1.2174±0.747	46.70±1.694/1.5522±1.053	58.35±38.10%	716,778	43,485,696	2,150,334	3× full	2,150,334
	FedSPD	56.48±1.786/1.3249±0.931	37.09±2.000/1.6815±1.175	55.01±53.30%	716,778	43,485,696	2,150,334	1× full	716,778
	MCFL	59.41±1.389/1.2463±0.808	46.54±1.817/1.5685±1.022	56.46±47.20%	716,778	43,485,696	2,150,334	2× full	1,433,556
	pFedCAM	62.86±1.511/1.1282±0.869	50.24±1.756/1.5018±0.992	60.73±44.20%	716,778	43,485,696	2,150,334	1× full	716,778
	MiB-SCFL	68.13±1.328/0.9183±0.656	54.84±1.603/1.2810±0.961	68.15±35.00%	716,778	43,485,696	2,150,334	$\approx 1\times$ full	$\approx 716,778$
HAR & $\epsilon = 0.03$	FedAvg	88.62±1.318/0.2788±0.458	76.64±1.614/0.5452±0.547	88.15±33.90%	186,310	184,960	186,310	1× full	186,310
	FedProx	88.56±1.258/0.2780±0.428	76.69±1.525/0.5388±0.665	88.10±30.90%	186,310	184,960	186,310	1× full	186,310
	CFL	83.71±1.674/0.5195±0.992	63.50±1.881/1.8612±1.436	83.10±63.60%	186,310	184,960	558,930	1× full	186,310
	B-CFL	88.52±1.644/0.3864±0.725	74.68±1.852/0.9807±1.199	87.96±57.60%	186,310	184,960	558,930	3× full	558,930
	FedMTL	55.39±1.822/1.0982±1.051	32.42±2.000/1.5155±1.081	45.52±102.10%	186,310	184,960	2,794,650	1× full	186,310
	FedRFC	93.79±1.110/0.1776±0.398	89.11±1.407/0.2689±0.903	93.69±25.00%	186,310	184,960	558,930	3× full	558,930
	FedSoft	90.43±1.347/0.2405±0.606	79.03±1.792/0.5529±1.140	90.31±36.90%	186,310	184,960	558,930	3× full	558,930
	FedSPD	81.83±1.703/0.5392±0.843	63.77±1.763/1.6622±1.377	79.20±69.50%	186,310	184,960	558,930	1× full	186,310
	MCFL	90.53±1.496/0.3489±0.873	72.47±1.941/1.5491±1.585	90.33±51.70%	186,310	184,960	558,930	2× full	372,620
	SC-FL	89.53±1.466/0.3626±0.784	75.76±1.911/0.8198±1.169	89.04±48.70%	186,310	184,960	558,930	3× full	558,930
	pFedCAM	85.53±1.733/0.5171±0.962	54.12±1.970/1.8168±1.555	84.13±93.20%	186,310	184,960	558,930	1× full	186,310
	MiB-SCFL	92.51±1.229/0.2156±0.754	89.09±1.288/0.2977±0.814	92.36±28.00%	186,310	184,960	558,930	$\approx 0.99\times$ full	$\approx 184,447$
PAMAP2 & $\epsilon = 0.07$	FedAvg	96.49±0.584/0.1164±0.218	89.73±0.813/0.2985±0.309	93.97±6.50%	159,314	7,391,488	159,314	1× full	159,314
	FedProx	96.46±0.599/0.1159±0.203	90.11±0.767/0.2886±0.340	93.97±5.00%	159,314	7,391,488	159,314	1× full	159,314
	CFL	96.16±0.630/0.1738±0.264	91.25±0.858/0.2867±0.386	93.79±12.60%	159,314	7,391,488	477,942	1× full	159,314
	B-CFL	96.32±0.828/0.1985±0.279	90.49±0.904/0.3278±0.508	93.89±23.30%	159,314	7,391,488	477,942	3× full	477,942
	FedMTL	79.84±0.950/0.9610±0.538	27.56±0.935/2.515±0.874	88.51±24.80%	159,314	7,391,488	2,389,710	1× full	159,314
	FedRFC	96.12±0.614/0.1675±0.325	85.36±0.919/0.4729±0.523	93.63±17.20%	159,314	7,391,488	477,942	3× full	477,942
	FedSoft	96.63±0.675/0.1766±0.462	92.20±0.660/0.5121±0.370	94.28±8.10%	159,314	7,391,488	477,942	3× full	477,942
	FedSPD	96.47±0.721/0.1821±0.447	92.77±0.782/0.4125±0.416	94.09±14.20%	159,314	7,391,488	477,942	1× full	159,314
	MCFL	96.71±0.691/0.1775±0.492	93.34±0.736/0.4998±0.431	94.33±9.60%	159,314	7,391,488	477,942	2× full	318,628
	SC-FL	96.59±0.752/0.1765±0.477	92.01±0.797/0.5130±0.553	94.25±11.10%	159,314	7,391,488	477,942	3× full	477,942
	pFedCAM	96.46±0.843/0.1703±0.355	90.30±0.889/0.3363±0.569	94.10±18.70%	159,314	7,391,488	477,942	1× full	159,314
	MiB-SCFL	96.77±0.706/0.1661±0.294	93.12±0.645/0.4062±0.401	94.83±15.70%	159,314	7,391,488	477,942	$\approx 0.99\times$ full	$\approx 157,721$
Shakespeare	FedAvg	50.39±3.844/1.8021±2.920	21.49±4.711/2.8200±3.267	50.42±228.40%	243,969	18,358,400	243,969	1× full	243,969
	FedProx	50.39±3.787/1.8021±2.747	21.49±4.769/2.8200±3.209	50.46±211.10%	243,969	18,358,400	243,969	1× full	243,969
	CFL	50.37±4.249/1.7829±3.093	20.82±5.000/2.8900±3.671	50.52±240.00%	243,969	18,358,400	731,907	1× full	243,969
	B-CFL	50.46±4.307/1.7588±3.036	20.00±4.942/2.8600±3.556	50.78±222.70%	243,969	18,358,400	731,907	3× full	731,907
	FedRFC	29.23±4.538/5.7300±4.364	14.29±4.480/3.1314±3.613	50.98±245.80%	243,969	18,358,400	731,907	3× full	731,907
	FedSoft	51.21±4.133/1.6887±2.978	24.00±4.884/2.7106±3.498	50.98±234.20%	243,969	18,358,400	731,907	3× full	731,907
	FedSPD	50.64±3.902/1.7300±2.689	22.31±4.827/2.7849±3.151	49.99±216.90%	243,969	18,358,400	731,907	1× full	243,969
	MCFL	50.11±4.018/1.7283±2.631	22.65±4.653/2.7966±3.382	50.02±205.30%	243,969	18,358,400	731,907	1× full	243,969
	SC-FL	50.94±3.960/1.7100±2.573	22.85±4.422/2.7851±3.440	50.72±199.60%	243,969	18,358,400	731,907	2× full	487,938
	pFedCAM	48.89±4.076/1.8493±2.804	18.16±4.596/3.0771±3.324	48.58±193.80%	243,969	18,358,400	731,907	1× full	243,969
	MiB-SCFL	50.13±3.729/1.7422±2.516	22.85±4.191/2.7689±2.862	50.10±188.00%	243,969	18,358,400	731,907	$\approx 1.15\times$ full	$\approx 280,564$